Gamma-ray spectroscopy with Ge(Li) and Abundance of Uranium Isotopes

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Abstract

To determine the isotope ratio of ^{235}U to ^{238}U in a uranium sample, a Ge(Li) detector was employed for gamma-ray spectroscopy, chosen for its high energy resolution. For energy calibration, ^{22}Na, ^{60}Co, and ^{137}Cs samples were measured for 5 minutes each, and the ^{152}Eu sample was measured for 50 minutes to determine energy efficiency. The uranium sample's energy profile was then measured over 50 minutes to assess the isotope ratio. Peak information from these energy profiles was extracted, energy calibration was applied, and energy efficiency was calculated. The isotope ratio was determined to be 0.3142(135) %. Given that the natural ratio is 0.7257(6) %, it was concluded that this sample source is depleted in ^{235}U.

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I. INTRODUCTION

Isotopic abundance means the relative proportion of each isotope of an element found in a natural sample. The atoms in the same element can have different numbers of neutrons, so the different isotopes have different atomic masses but identical chemical properties due to the same number of protons. The abundance of these isotopes in nature is expressed as a percentage or fraction of the atom number of the certain isotope over that of the element.

Isotopic abundance is important in several fields, such as nuclear physics, chemistry, and geology. In nuclear physics, it is used to determine the radioactive properties of the isotopes, as different isotopes can have different decay modes and half-lives, because the stability of the isotopes varies. Isotopic abundance measurements are typically conducted by using techniques such as mass spectrometry or gamma spectroscopy, depending on the type of isotope.

Understanding isotopic abundance is key to calculating beam activity accurately and identifying the source in experiments. For instance, when using isotopic sources like ^{60}Co or ^{137}Cs, knowing their isotopic composition ensures the correct interpretation of gamma-ray spectra. Different isotopes can produce different gamma-ray peaks, and their relative intensities can help to distinguish one isotope from another. Therefore, isotopic abundance provides insight into the proportion of radioactive material and helps quantify nuclear properties during experiments.

II. MATERIALS AND EXPERIMENTAL PROCEDURES

The application of semiconductor detectors in gamma-ray detection has led to substantial improvements in measurement capabilities. Compared to these, gamma-ray peaks obtained with NaI detectors are relatively broad, making it difficult to resolve two closely spaced peaks, and low-energy peaks may not be easily detectable. Semiconductor detectors made from lithium-compensated germanium or silicon offer significantly enhanced energy resolution.

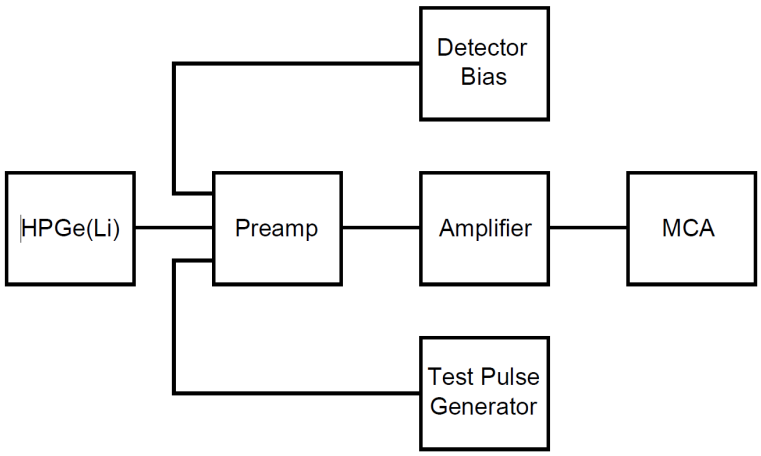


FIG. 1: Major components of the gamma-ray spectroscopy system set up for this experiment.[1]

The Germanium detector should only be powered when cold. The detector is cooled by a Nitrogen bath contained in a large dewar. Once the dewar is filled, cooling the Ge detector fully takes about 6 hours. Please ensure with instructor that the detector can be powered before operating it.

In the procedure, we always need to power up and down slowly. While observing the raw signal on the oscilloscope, as we raise the high voltage(HV), the signal will get large and positive and settle down to zero. Increase the HV by small steps such as 200 V and wait for the signal to return to zero before increase the HV. Repeat these steps until reach 1500 V. And do not power over 1500 V. After finishing the experiment, power the HV all the way down with the small steps, and wait a few minutes to turn the HV off.

First, we measured the sources to find the appropriate setting of the amplifier. While controling the the knobs of the amplifier, we checked that 1408.013(3) keV peak for ^{152}Eu source would be on the range of measurements. The appropriate setting of the amplifier is 20 on the coarse gain, and 12 cycle with 40 and 1/5 on the fine gain. We measured the decay profile during 5 minutes from ^{22}Na, ^{60}Co, and ^{137}Cs samples for the purpose of energy calibration. Also, we measured the decay profile during 50 minutes from ^{152}Eu sample for the purpose to get relative energy efficiency. Finally, we measured the decay profile during 50 minutes from the uranium mixture sample to figure out the isotope ratio.

III. RESULTS AND DISCUSSION

A. Extraction of the Peaks from the Samples

TABLE I: The summary of the analysis of the peaks from the samples, 2 peaks from ^{22}Na and ^{60}Co, and 1 peak from ^{137}Cs. RMS stands for the Root Mean Square Error, and SEM stands for the Standard Error of the Mean. Backgrounds were ignored due to the small amount.

Peaks	N_{Total}	Mean (Bin)	RMS (Bin)	SEM (Bin)
^{22}Na				
1st	4,289	2,874.357	6.599	0.101
2nd	910	6,964.830	5.186	0.172
^{60}Co				
1st	3,132	6,422.551	5.317	0.095
2nd	2,633	7,275.399	5.431	0.106
^{137}Cs				
1st	3,733	3,681.460	3.898	0.064

TABLE II: The summary of the analysis of the peaks from the samples. 8 peaks from ^{152}Eu, 5 peaks from ^{235}U, and 2 peaks from ^{238}U.

Peaks	N_{Total}	$N_{\text{Background}}$	N_{Signal}	Mean (Bin)	RMS (Bin)	SEM (Bin)
^{152}Eu						
1st	53,965	4,948	49,017	779.127	2.502	0.011
2nd	19,231	1,406	17,825	1,979.210	2.882	0.022
3rd	2,928	91	2,837	7,677.351	5.671	0.106
4th	3,517	602	2,915	5,300.250	4.360	0.081
5th	2,928	595	2,333	6,092.611	4.978	0.103
6th	4,177	991	3,186	4,308.581	3.850	0.068
7th	2,392	519	1,873	5,952.916	4.904	0.113
8th	9,405	1,958	7,447	1,444.186	2.653	0.031
^{235}U						
1st	36,439	6,034	30,405	1127.480	2.499	0.014
2nd	10,459	6,225	4,234	899.989	2.459	0.038
3rd	8,175	5,707	2,468	1,006.129	2.587	0.052
4th	5,442	2,768	2,674	1,233.144	2.567	0.050
5th	2,454	1,967	487	1,215.628	1.804	0.082
^{238}U						
1st	3,105	143	2,962	5,499.681	4.301	0.079
2nd	1,807	454	1,353	4,243.000	3.640	0.099

First, all the information about the peaks in the energy profiles from the samples are presented in the TABLE I and the TABLE II. The background signals for the peaks were approximated by linear equations with appropriate processing respectively.

B. Linear Fitting and Energy Calibration

The linear fitting of the data and the error would be

$$\begin{aligned} f(x) &= \alpha x - \beta \\ \sigma_f^2(x) &= x^2\sigma_\alpha^2 + \sigma_\beta^2 + 2x\mathrm{cov}(\alpha, \beta) \end{aligned} \tag{1}$$

and the calibration for the another data point (x_j, y_j, σ_j) would be

$$\begin{aligned} f(x_j) &= \alpha x_j - \beta \\ \sigma_T^2(x_j) &= \alpha\sigma_j^2 + \sigma_f^2(x_j) \end{aligned} \tag{2}$$

We can try to use 5 peaks from the ^{22}Na, ^{60}Co, and ^{137}Cs for linear fitting and the calibration for the energy. Or we can use 8 peaks from the ^{152}Eu, or we can use all 13 peaks simultaneously. The coefficients for the linear fitting for the three cases are like below.

$\alpha_{E_5} = 0.1867, \beta_{E_5} = -25.50, \sigma^2_{\alpha_{E_5}} = 2.152\times10^{-11}, \mathrm{cov}(\alpha_{E_5}, \beta_{E_5}) = -1.038\times10^{-7}, \sigma^2_{\beta_{E_5}} = 0.0005629$

$\alpha_{E_8} = 0.1862, \beta_{E_8} = -23.54, \sigma^2_{\alpha_{E_8}} = 2.691\times10^{-12}, \mathrm{cov}(\alpha_{E_8}, \beta_{E_8}) = -3.415\times10^{-9}, \sigma^2_{\beta_{E_8}} = 7.432\times10^{-6}$

$\alpha_{E_{13}} = 0.1863, \beta_{E_{13}} = -23.60, \sigma^2_{\alpha_{E_{13}}} = 1.637\times10^{-12}, \mathrm{cov}(\alpha_{E_{13}}, \beta_{E_{13}}) = -2.353\times10^{-9}, \sigma^2_{\beta_{E_{13}}} = 6.334\times10^{-6}$

TABLE III is showing the calibration results from the three method of the linear fitting.

The minimum of $\sum(E - E_{ref})^2$ is given by E_5. It is because the bin size for the energy is too large to get the accurate energy values for the peaks. Some of the SEM values for the ^{152}Eu are small because of the long measurement time. Paradoxically, the tolerance for the error has been reduced by the longer measurement time, but the error from the true value still exists due to systematic limitations, so it can be said that the fitting results are becoming more inaccurate if we use the information from the ^{152}Eu for the linear fitting. So we need to use the less time measurement results for our linear fitting. So, it is appropriate to use the coefficients of E_5 for the processes from now on, so, $E_{fit} = E_5$. FIG. 2 is showing the energy residuals from the calibration using E_5.

TABLE III: The summary of the energy calibration results. All unit of energies is keV. Red means negative amount.

Peaks	E_{ref}[2]	$E_5 =$ $E_{\text{fit with 5 peaks}}$	$E_5 - E_{ref}$	$E_8 =$ $E_{\text{fit with 8 peaks}}$	$E_8 - E_{ref}$	$E_{13} =$ $E_{\text{fit with 13 peaks}}$	$E_{13} - E_{ref}$
^{22}Na							
1st	511.000(1)	511.0038(120)	0.0038(121)	511.7077(32)	0.7077(33)	511.7817(26)	0.7817(27)
2nd	1274.537(3)	1274.4937(128)	0.0433(131)	1273.4077(96)	1.1293(101)	1273.6723(74)	0.8647(80)
^{60}Co							
1st	1173.228(3)	1173.2769(109)	0.0489(113)	1172.4282(87)	0.7998(92)	1172.6676(67)	0.5604(74)
2nd	1332.492(4)	1332.4616(140)	0.0304(145)	1331.2398(102)	1.2522(109)	1331.5189(79)	0.9731(88)
^{137}Cs							
1st	661.657(3)	661.6502(96)	0.0068(101)	662.0009(45)	0.3439(54)	662.1125(36)	0.4555(47)
^{152}Eu							
1st	121.7817(3)	119.9274(204)	1.8543(204)	121.5480(19)	0.2337(20)	121.5244(19)	0.2573(19)
2nd	344.2785(12)	343.9239(154)	0.3546(154)	345.0194(22)	0.7409(25)	345.0517(19)	0.7732(23)
3rd	1408.013(3)	1407.4863(155)	0.5267(158)	1406.0886(107)	1.9244(111)	1406.3864(83)	1.6266(88)
4th	964.072(18)	963.7986(113)	0.2734(213)	963.4410(82)	0.6310(208)	963.6280(94)	0.4440(203)
5th	1112.076(3)	1111.6933(100)	0.3827(104)	1110.9890(82)	1.0870(87)	1111.2130(63)	0.8630(70)
6th	778.9045(24)	778.7028(83)	0.2017(87)	778.7791(90)	0.1254(59)	778.9200(42)	0.0155(48)
7th	1085.837(10)	1085.6192(104)	0.2178(145)	1084.9761(90)	0.8609(134)	1085.1935(74)	0.6435(124)
8th	244.6974(8)	244.0613(176)	0.6361(176)	245.3910(18)	0.6936(20)	245.3983(18)	0.7009(19)
		$\sum (E_5 - E_{ref})^2$	4.5608	$\sum (E_8 - E_{ref})^2$	11.2264	$\sum (E_{13} - E_{ref})^2$	7.9846

C. Energy Resolution and Energy Efficiency

In FIG. 3, we can see the noticeably large peak in the energy resolution at 511 keV from ^{22}Na sample. This 511 keV photons produced by an electron interacting with a positron, which we calls annihilation radiation. The annihilation radiation differs from monoenergetic gamma rays emitted during radioactive decay due to its production mechanism, which causes Doppler broadening. [3] This results in the annihilation peak in the photon spectrum having a larger full width at half maximum (FWHM) than gamma rays from decay processes.

The relative energy efficiency ε, depends on the energy, should be calculated to get the isotope ratio. ε can be calculated like this. First, select the reference peak and consider ε of that energy would be 1. Then the expected number of dectected particle at the peak other than reference peak would be $N_i^{Expected} = \frac{N_{ref}}{I_{ref}} I_i$. Then

$$\varepsilon_i = \frac{N_i}{N_i^{Expected}} = \frac{N_i I_{ref}}{N_{ref} I_i} \tag{3}$$

The ε for the peaks for ^{152}Eu are calculated in TABLE IV, and by fitting linearly, we can calculate ε for the peaks for the uranium sample.

The coefficients for the linear fitting, which depend on the energies, are introduced below.

For 119.9274 keV $< E <$ 244.0613 keV, the linear fitting coefficients for ε are $\alpha_\varepsilon = -0.003451, \beta_\varepsilon = 1.414$, $\sigma^2_{\alpha_\varepsilon} = 1.139 \times 10^{-8}, \text{cov}(\alpha_\varepsilon, \beta_\varepsilon) = -2.098 \times 10^{-6}$, and $\sigma_{\beta^2_\varepsilon} =$

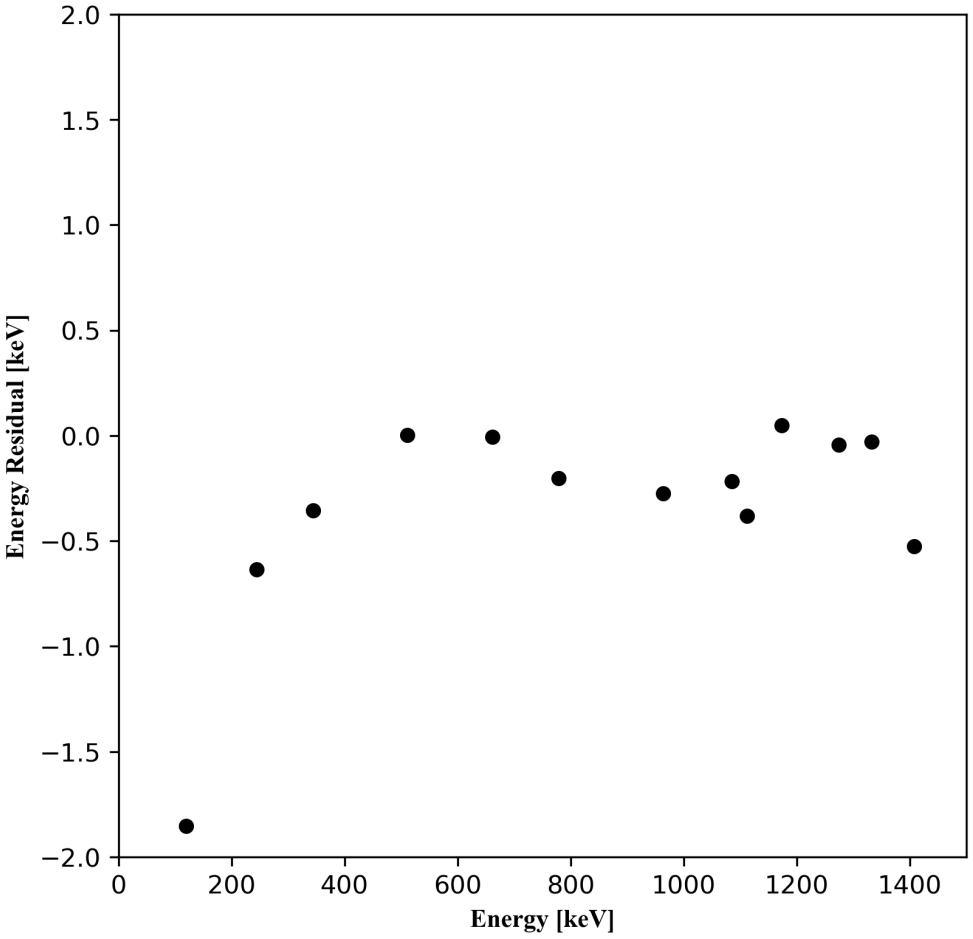


FIG. 2: The energy residual from E_5.

0.0004303.

For 343.9239 keV $< E <$ 778.7028 keV, the linear fitting coefficients for ε are $\alpha_\varepsilon = -0.0005663, \beta_\varepsilon = 0.5834$, $\sigma^2_{\alpha_\varepsilon} = 1.513 \times 10^{-10}, \mathrm{cov}\,(\alpha_\varepsilon, \beta_\varepsilon) = -9.193 \times 10^{-8}$, and $\sigma^2_{\beta_\varepsilon} = 6.270 \times 10^{-5}$.

For 963.7986 keV $< E <$ 1085.6192 keV, the linear fitting coefficients for ε are $\alpha_\varepsilon = -6.895 \times 10^{-5}, \beta_\varepsilon = 0.182$, $\sigma^2_{\alpha_\varepsilon} = 1.158 \times 10^{-9}, \mathrm{cov}\,(\alpha_\varepsilon, \beta_\varepsilon) = -1.174 \times 10^{-6}$, and $\sigma^2_{\beta_\varepsilon} = 0.001194$.

The linear fitting result is shown in the FIG. 4. From that result, ε for the peaks of uranium sample was calculated, and the results are in the TABLE V.

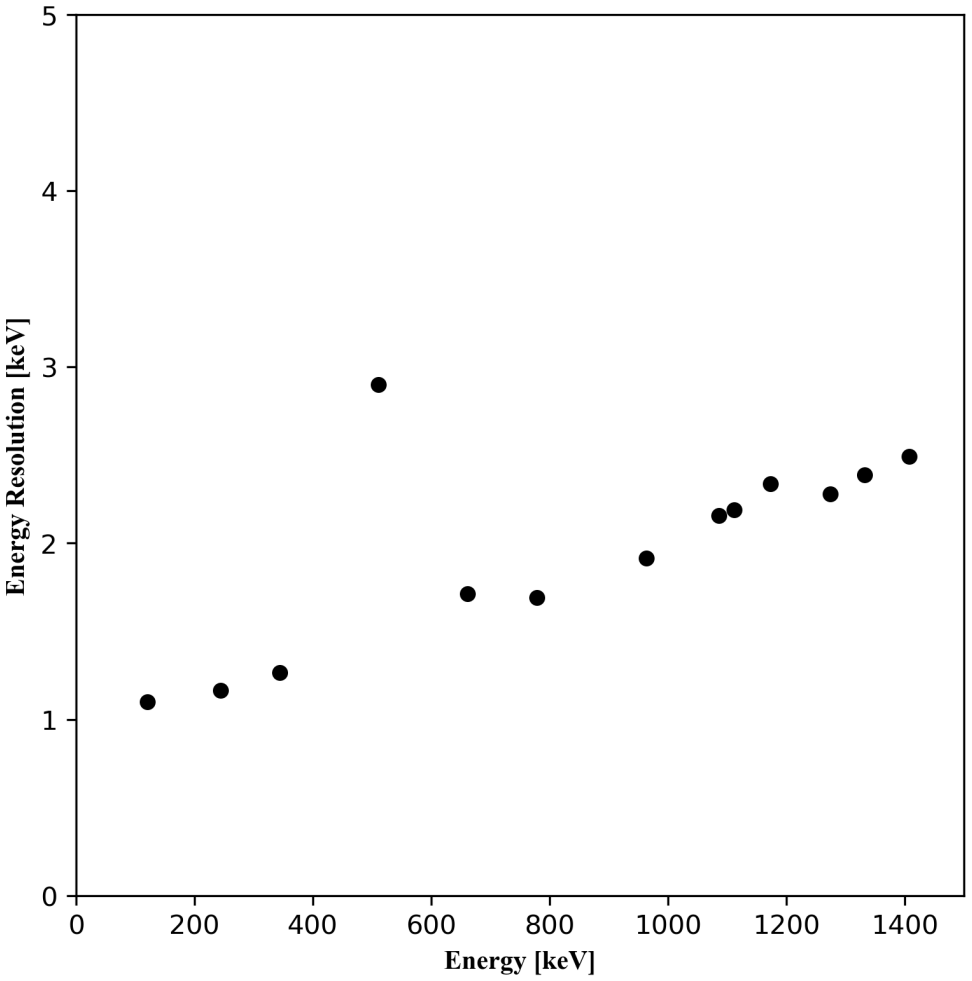


FIG. 3: The energy resolution. The energy resolution of 511 keV from ^{22}Na is noticeably large.

TABLE IV: The relative energy efficiency(ε) calculation from the ^{152}Eu data.

Peaks	E_{fit} (keV)	P_{ref}[2]	N_{Peak}	N_{Expected}	ε
^{152}Eu					
1st	119.9274(204)	0.2841(13)	49,017(243)	49,017(399)	1.0000(95)
2nd	343.9239(154)	0.2658(12)	17,825(144)	45,860(372)	0.3887(44)
3rd	1407.4863(155)	0.2085(9)	2,837(55)	35,973(288)	0.0789(17)
4th	963.7986(113)	0.1462(6)	2,915(64)	25,225(199)	0.1156(27)
5th	1111.6933(100)	0.1340(6)	2,333(59)	23,120(187)	0.1009(27)
6th	778.7028(83)	0.1296(6)	3,186(72)	22,361(183)	0.1425(34)
7th	1085.6192(104)	0.1013(6)	1,873(54)	17,478(157)	0.1072(32)
8th	244.0613(176)	0.0755(4)	7,447(107)	13,026(112)	0.5717(95)

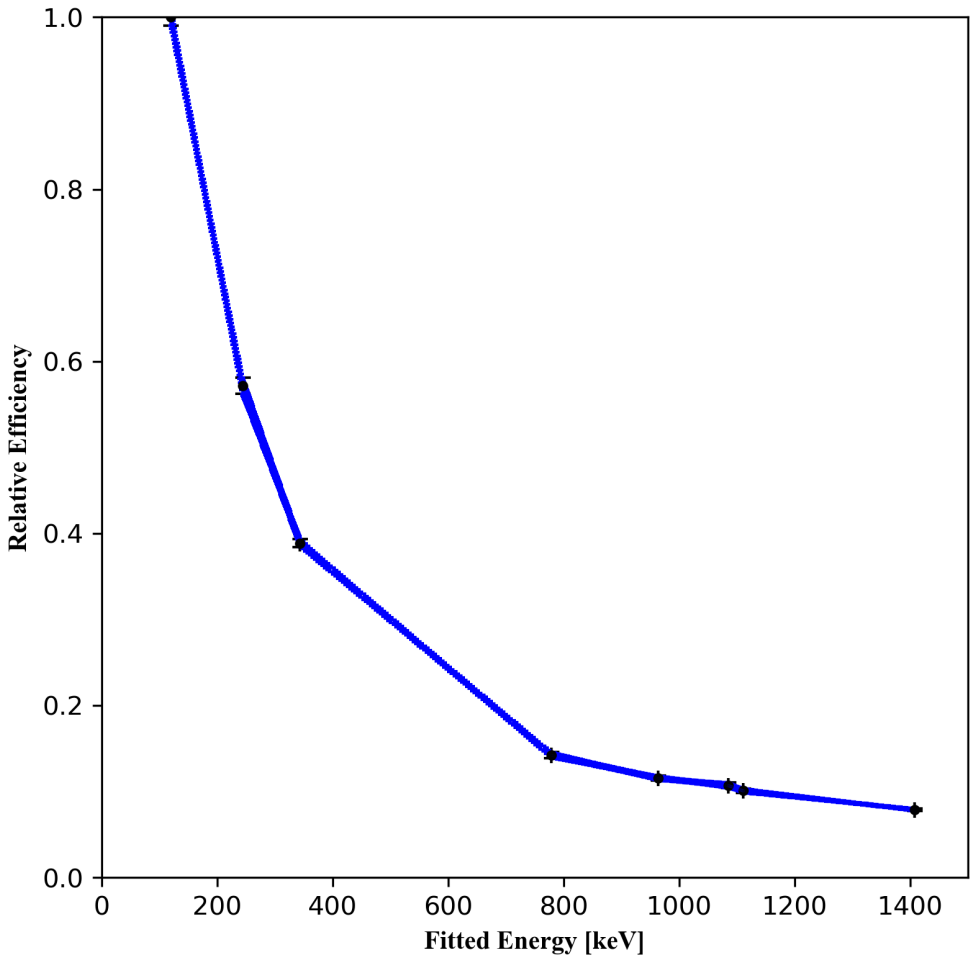


FIG. 4: The linearly fitted relative energy efficiency. The black dots and the errorbars are from the TABLE IV. The blue line indicates the guessed values at the energies.

TABLE V: The relative energy efficiency(ε) for the uranium sample.

Peaks	P_{ref}	E_{ref} (keV)	E_{fit} (keV)	ε
^{235}U [4]				
1st	0.572(5)	185.715(5)	184.9477(190)	0.7756(68)
2nd	0.1096(8)	143.76(2)	142.4864(216)	0.9222(80)
3rd	0.0508(4)	163.33(2)	162.2976(212)	0.8538(71)
4th	0.0501(5)	205.311(10)	204.6700(189)	0.7076(72)
5th	0.0108(2)	202.11(2)	201.4007(204)	0.7189(71)
^{238}U [5]				
1st	0.00842(8)	1,001.03(10)	1,001.0225(440)	0.1494(33)
2nd	0.00317(5)	766.42(10)	766.4620(440)	0.1130(21)

D. Abundance of Uranium Isotopes and Isotope Ratio

The decay constant λ is an inverse of half-life $t_{1/2}$.

$$\lambda = \frac{\ln(2)}{t_{1/2}} \tag{4}$$

An exponential decay can be described like this.

$$\begin{aligned} N(t) &= N_0 e^{-\lambda t} \\ \frac{dN(t)}{dt} &= -\lambda N_0 e^{-\lambda t} = -\lambda N(t) \end{aligned} \tag{5}$$

where N_0 is the initial quantity of the material that will decay. What we are detecting is the time integration of the decaying particles.

$$\int_0^{\Delta t} \frac{dN(t)}{dt} = \left[N_0 e^{-\lambda t}\right]_0^{\Delta t} = N_0 \left(e^{-\lambda \Delta t} - 1\right) \approx -\lambda N_0 \Delta t \tag{6}$$

The decayed number of particles according to the energy E_i, $N_{d,i} = \lambda P_i N_0 \Delta t$ would be detected, where P_i is the intensity at the peak i. We want to obtain $\frac{N_{0,\mathrm{U}235}}{N_{0,\mathrm{U}238}}$, and $\frac{N_{d,i,\mathrm{U}235}/P_i}{N_{d,j\mathrm{U}238}/P_j} = \frac{\lambda_{\mathrm{U}235} N_{0,\mathrm{U}235} \Delta t}{\lambda_{\mathrm{U}238} N_{0,\mathrm{U}238} \Delta t}$, so $\frac{N_{0,\mathrm{U}235}}{N_{0,\mathrm{U}238}} = \left(\frac{1}{N_i}\sum_i \frac{N_{d,i,\mathrm{U}235}}{\lambda_{\mathrm{U}235} P_i}\right) / \left(\frac{1}{N_j}\sum_j \frac{N_{d,j,\mathrm{U}238}}{\lambda_{\mathrm{U}238} P_j}\right)$. The calculation results are suggested in TABLE VI.

TABLE VI: The calculation of the isotope ratio between ^{235}U and ^{238}U.

Peaks	N_{Signal}	ε	$N_d = N_{\mathrm{Signal}}/\varepsilon$	N_d/λ (y)	$N_{\mathrm{Decay}} = N_d/(\lambda P_{ref})$ (y)
^{235}U, $t_{1/2} = 7.040(10) \times 10^8$ y [6]					
1st	30,405(206)	0.7756(68)	39,199(432)	$3.981(44) \times 10^{13}$	$6.960(98) \times 10^{13}$
2nd	4,234(129)	0.9222(80)	4,592(146)	$4.664(148) \times 10^{12}$	$4.255(139) \times 10^{13}$
3rd	2,468(118)	0.8538(71)	2,890(140)	$2.935(142) \times 10^{12}$	$5.778(284) \times 10^{13}$
4th	2,674(91)	0.7076(72)	3,779(134)	$3.838(136) \times 10^{12}$	$7.661(282) \times 10^{13}$
5th	487(66)	0.7189(71)	677(93)	$6.874(942) \times 10^{11}$	$6.365(880) \times 10^{13}$
Mean				$\frac{1}{5}\sum_i \frac{N_{d,i,\mathrm{U}235}}{\lambda_{\mathrm{U}235} P_i}$	$6.204(196) \times 10^{13}$
^{238}U, $t_{1/2} = 4.463(5) \times 10^9$ y [5]					
1st	2,962(57)	0.1494(33)	19,824(583)	$1.276(38) \times 10^{14}$	$1.516(47) \times 10^{16}$
2nd	1,353(48)	0.1130(21)	11,977(477)	$7.711(307) \times 10^{13}$	$2.433(104) \times 10^{16}$
Mean				$\frac{1}{2}\sum_j \frac{N_{d,j,\mathrm{U}238}}{\lambda_{\mathrm{U}238} P_j}$	$1.974(57) \times 10^{16}$

The relative abundance of ^{235}U for ^{238}U is $\frac{6.204(196)\times 10^{13}\ \mathrm{y}}{1.974(57)\times 10^{16}\ \mathrm{y}} = 0.3142(135)$ %. The natural abundance of ^{235}U is 0.7204(6) % and that of ^{238}U is 99.2742(10) %[7]. The natural ratio of them is 0.7257(6) %, so the sample source of the experiment is depleted of ^{235}U.

IV. CONCLUSION

To obtain the isotope ratio between ^{235}U and ^{238}U in the uranium sample, the Ge(Li) detector was used for Gamma-ray spectroscopy for high energy resolution. The measurement time was 5 minutes for ^{22}Na, ^{60}Co, and ^{137}Cs samples for the energy calibration and 50

minutes for the ^{152}Eu sample to obtain the energy efficiency. To obtain the isotope ratio, we measured energy profile for 50 minutes for the uranium sample. The information of the peaks from the energy profiles was extracted. The energy calibration was conducted, and the energy efficiency was obtained. The isotope ratio was calculated as 0.3142(135) %. The natural ratio is 0.7257(6) %, so we found this sample source is depleted of ^{235}U.

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